

Solving linear systems (contd.)

(Sect 1.3 & 1.4 of the Notes)

Last time:

Easy to solve if system (corresponding matrix) is in row echelon form

$$\left(\begin{array}{cccc|c} 2 & 1 & 3 & 5 & 5 \\ 0 & 1 & 7 & 5 & 4 \\ 0 & 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & 6 & 2 \end{array} \right)$$

row echelon form

unique solution

$$\left(\begin{array}{cccccc|c} 2 & 1 & 4 & 7 & 5 & 6 & 7 & 1 \\ 0 & 0 & 2 & 1 & 4 & 9 & 5 & 4 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

row echelon form

infinitely many solutions

Can see at one glance how many solutions.

The set of solutions is also relatively easy to determine.

But usually, the system is not in row echelon form.



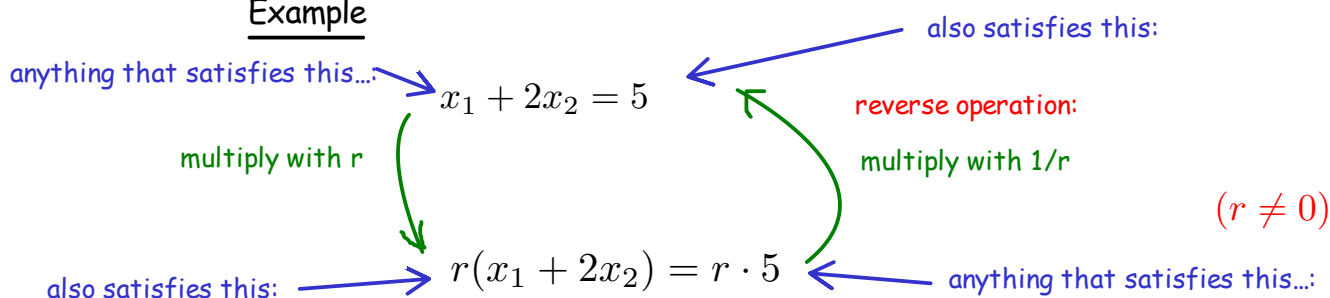
We want that A and A' have the same solution set:

- * Each solution for A must be a solution for A' ,
- * Each solution for A' must be a solution for A .

3 operations that don't change the solution set:

- 1.) interchanging two rows ($R_i \leftrightarrow R_j$)
- 2.) multiplying a row by a number $r \neq 0$ ($R_i \rightarrow rR_i$)

Example



3.) Add a multiple of a row to another row ($R_i \rightarrow R_i + rR_j$)

Example $x_1 + 2x_2 = 5$ $R_1 \rightarrow R_1 + 10R_2$ (1)

$$2x_1 + x_2 = 7$$

We get

$$x_1 + 2x_2 + 10(2x_1 + x_2) = 5 + 10 \cdot 7$$

$$2x_1 + x_2 = 7$$

(2)

Obvious: Any solution for (1) is also solution for (2).

The operation " $R_1 \rightarrow R_1 + 10R_2$ " has a reverse operation $R_1 \rightarrow R_1 - 10R_2$

Using $R_1 \rightarrow R_1 - 10R_2$, we get back to

$$x_1 + 2x_2 = 5$$

$$2x_1 + x_2 = 7$$

The same argument as above shows that any solution for (2) is also a solution for (1).

We can apply these three elementary operation as often as we like. We will not change the solution set.

Now we want to show that these three operations are sufficient to turn any given matrix into row echelon form.

Example $\left(\begin{array}{ccccc|c} 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 5 & 5 & 9 \\ 2 & 4 & 6 & 8 & 8 & 5 \\ 3 & 6 & 8 & 10 & 10 & 7 \end{array} \right)$ $R_3 \rightarrow R_3 - 2R_1$
 $R_4 \rightarrow R_4 - 3R_1$

$$\left(\begin{array}{ccccc|c} 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 5 & 5 & 9 \\ 0 & 0 & 2 & 4 & 4 & 1 \\ 0 & 0 & 2 & 4 & 4 & 1 \end{array} \right) R_2 \leftrightarrow R_3$$

From now on, "ignore" the first row.

$$\left(\begin{array}{ccccc|c} 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 & 4 & 1 \\ 0 & 0 & 0 & 5 & 5 & 9 \\ 0 & 0 & 2 & 4 & 4 & 1 \end{array} \right) R_4 \rightarrow R_4 - R_2$$

free variables

x_1
 x_2
 x_3
 x_4
 x_5

$$\left(\begin{array}{ccccc|c} 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 & 4 & 1 \\ 0 & 0 & 0 & 5 & 5 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Row echelon form

$$\begin{array}{l} x_2 = s \\ \hline x_5 = t \end{array}$$

The third row tells us that

$$5x_4 + 5t = 9$$

Hence

$$x_4 = 1.8 - t$$

Similarly row 2 gives us

$$x_3 = -3.1$$

and row 1 tells us that

$$x_1 = 4.6 - 2s.$$

We write the set of solutions in vector form:

$$\left\{ \begin{pmatrix} 4.6 \\ 0 \\ -3.1 \\ 1.8 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}$$